

Uday Chandra Bhookya

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PROFILE SUMMARY

Applied AI Engineer with **4 years** of experience shipping production-grade **agentic systems** across defense, engineering, and commerce. Bridging a strong foundation in **classical ML** with advanced **multi-agent orchestration** (LangGraph, Google ADK, MCP), **RAG, fine-tuning, and evaluation pipelines**, specializing in architecting the end-to-end enterprise AI pipelines that take models from prototype to deployment. Co-author of **5 research papers** and **4 patents**, driven by solving complex customer problems with measurable ROI.

EDUCATION

Indian Institute of Information Technology, Tiruchirapalli
Bachelor of Technology in Computer Science and Engineering

India
Aug. 2018 – May 2022

EXPERIENCE

Data Science & GenAI Consultant
Deloitte

Mar 2026 – Present
Hyderabad, India

- **Architected** a secure **AI Document Classification Engine** for an **American defense company's** SAP migration, accelerating compliance reviews by **80%**. Built a **multi-stage agentic pipeline**, LLM classifier over complex unstructured formats, an **evaluator agent** and **reinforcement learning**. **Optimized throughput and inference cost** via batched async LLM calls and prompt caching, cutting end-to-end processing time by **2x**.
- **Designed and deployed** an end-to-end **Agentic Text-to-Engineering System** for a **civil construction company**, cutting diagram turnaround time by **70%**. Engineered a **multimodal multi-agent GenAI pipeline** (**vision models** + reasoning agents): generation of standards-compliant P&ID diagrams with CAD output, an **evaluator agent** with human-in-the-loop review, a **comparison agent** for semantic diff across versions, and a **procurement agent** surfacing material insights.
- **Architecting** (ongoing) an **Agentic Recommendation System** for an **autonomous commerce platform**, replacing static pipelines with **reasoning agents** that infer purchase intent from implicit signals (clicks, dwell, scroll, hover) in real time, plus a **conversational agent** for context-aware discovery moving beyond traditional filter-and-rank recommendations.

Data Scientist
Digitate – Tata Research

Aug 2022 – Mar 2026
Pune, India

- **Designed and led** a centralized **AI-driven incident resolution agent** using an agentic AI architecture with memory and context awareness, enabling intelligent troubleshooting, safer automated actions, and human-in-the-loop decision-making.
- **Scaled** this agent across **5+ products** and **10+ enterprise customers**, establishing a reusable architecture; owned end-to-end system design and collaborated across frontend and backend teams to deliver production-grade AI systems.
- **Architected and built** a cloud-native (**AWS + GCP**) **multi-agent AI platform** using **Google ADK, AWS Strands, MCP, and A2A protocols**, integrating ServiceNow, Gmail, and Slack to automate end-to-end IT incident workflows and reduce resolution time by **30%+**.
- **Developed** ML-driven **cloud cost optimization** solutions for Ignio's CloudOps, deployed across **5 enterprises**, delivering **\$6M+ savings** and **30–40% reduction** in annual cloud expenditure using forecasting, anomaly detection, and recommendation models.
- **Built** GenAI-powered features including **RAG** and **Table Q&A** systems, enabling natural language querying over structured and unstructured data and delivering concise, actionable insights for business users.
- **Architected** a **data storytelling platform** that auto-generated narratives from 50K+ insights, accelerating insight discovery by **90%** using two-tower models and graph-based ranking algorithms.
- **Developed** a **multilingual GenAI document assistant**, automating product documentation generation and enhancement, improving global accessibility and user engagement.
- **Built and deployed** a **TF-IDF based text classification** pipeline, achieving **85%+ precision** and streamlining large-scale text analytics workflows.
- **Enhanced** decision-tree models to derive interpretable and optimized classification rules, contributing to a **published research paper** on decision intelligence systems.
- **Provided** onboarding and L3-level support for **15+ customer engagements**, while leading and mentoring junior data scientists to improve delivery quality and system reliability.

- **Enhanced** key product pages across the platform to improve usability and deliver a seamless user experience, boosting overall customer satisfaction and engagement.
- **Developed** a utility for a coupon discount management system, enabling the product team to make data-driven decisions by streamlining analysis.

TECHNICAL SKILLS

ML / DL: Recommendation Systems, Time Series Analytics, Classification, Regression, CNNs, Predictive Modeling, Anomaly Detection, Hypothesis Testing, Pandas, NumPy, TensorFlow, PyTorch, Scikit-learn

Generative AI: Bedrock, Google ADK, Fine-Tuning LLMs, RAG, LangChain, LangGraph, Agentic AI, MCP, Multi-Agent Systems

Programming Languages: Python, Java, C, JavaScript

Big Data & Cloud: PySpark, GCP, Azure

Databases & Web: MySQL, PostgreSQL, Django, Flask, ReactJs

PUBLICATIONS

- **U. C. Bhookya**, R. Babu, S. Saluja, and M. Natu, “Data Stories from Insights for IT Operations,” *IEEE Big Data*, 2025. [Link]
- **U. C. Bhookya**, K. Jethuri, S. R. Ravuru, and M. Natu, “Addressing Spend Leakage and Optimization of Cloud Costs,” *IEEE Big Data*, 2024, Washington DC, USA. [Link]
- O. Rai, **U. C. Bhookya**, R. Babu, K. Jethuri, and M. Natu, “Cloud Cost Optimization with Data Storytelling,” *IEEE ICDMW*, 2025. [Link]
- S. N. Samudrala, **U. C. Bhookya**, and M. Natu, “Mining patterns for proactive management of procure-to-pay exceptions,” *IEEE ICDMW*, 2024, Abu Dhabi. [Link]
- S. Saluja, R. Babu, **U. C. Bhookya**, and M. Natu, “Addressing Insight Fatigue with Insight Summarization,” *COMSNETS*, 2025.
- **U. C. Bhookya**, “Statistical Significance Tests: A Statistical Way to Compare Data Populations,” *Digitate Blog*, Mar. 2023. [Link]

PROJECTS

AI Data Analyst Agent | *FastAPI, LangGraph, Python* [GitHub]

- **Built** a multi-agent AI data analyst using **FastAPI + LangGraph**, enabling automated insight generation and conversational querying over structured data; supports end-to-end analytics workflows.

Real-Time Hand Cricket: CNN-Based Multiplayer Game | *VGG16, Python, Flutter* [GitHub]

- **Built** a real-time gesture recognition system using a customized **VGG16 CNN model**, achieving **94% accuracy**; delivered as a mobile app in Flutter.

Domain-Specific LLM Fine-Tuning | *Python, HuggingFace, PyTorch*

- Fine-tuned an LLM for enhanced performance on specialized, domain-specific tasks using supervised fine-tuning techniques.

PATENTS

- **U.C. Bhookya**, K. Jethuri, S. R. Ravuru, and M. Natu. Methods and systems to optimize cloud cost by analyzing resource utilization. Patent Application No. 202421066668 (Patent pending).
- **U.C. Bhookya**, K. Jethuri, S. R. Ravuru, and M. Natu. Methods and systems to optimize cloud cost by analyzing pricing models. Patent Application No. 202421066667 (Patent pending).
- **U.C. Bhookya**, K. Jethuri, S. R. Ravuru, and M. Natu. Methods and systems to optimize cloud cost by analyzing cloud resource usage. Patent Application No. 202421066669 (Patent pending).
- R. Babu, **U.C. Bhookya**, and M. Natu. Data driven insight generation and creation of contextually consistent chains thereof. Patent Application No. 202421093804 (Patent pending).